

Problem

Find the Fourier transforms of:

(a) $2 \cos 2tu(t)$

(b) $0.5 \sin 10tu(t)$

Step-by-step solution

Step 1 of 4

(a)

Consider the following function:

$$H(t) = 2 \cos 2tu(t) \dots\dots (1)$$

It is known that,

$$\begin{aligned} H(\omega) &= F(\cos \omega_0 t f(t)) \\ &= \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)] \end{aligned}$$

Here,

$$\omega_0 = 2$$

And,

$$\begin{aligned} F(\omega) &= F[u(t)] \\ &= \pi \delta(\omega) + \frac{1}{j\omega} \end{aligned}$$

Step 2 of 4

Therefore, the Fourier transform of equation (1).

$$\begin{aligned}
 H(\omega) &= 2 \times \frac{1}{2} \left[\pi \delta(\omega + 2) + \frac{1}{j(\omega + 2)} + \pi \delta(\omega - 2) + \frac{1}{j(\omega - 2)} \right] \\
 &= \left[\pi \delta(\omega + 2) + \frac{1}{j(\omega + 2)} + \pi \delta(\omega - 2) + \frac{1}{j(\omega - 2)} \right] \\
 &= \pi [\delta(\omega + 2) + \delta(\omega - 2)] + \left[\frac{1}{j(\omega + 2)} + \frac{1}{j(\omega - 2)} \right] \\
 &= \pi [\delta(\omega + 2) + \delta(\omega - 2)] + \left[\frac{j(\omega - 2) + j(\omega + 2)}{j(\omega + 2)j(\omega - 2)} \right] \\
 &= \pi [\delta(\omega + 2) + \delta(\omega - 2)] - j \left[\frac{(\omega - 2) + (\omega + 2)}{(\omega + 2)(\omega - 2)} \right] \\
 &= \pi [\delta(\omega + 2) + \delta(\omega - 2)] - j \left[\frac{2\omega}{(\omega + 2)(\omega - 2)} \right] \\
 H(\omega) &= \pi [\delta(\omega + 2) + \delta(\omega - 2)] - j \left[\frac{2\omega}{(\omega^2 - 4)} \right]
 \end{aligned}$$

Therefore, the Fourier Transform of $2 \cos 2t u(t)$ is

$$\boxed{H(\omega) = \pi [\delta(\omega + 2) + \pi \delta(\omega - 2)] - j \left[\frac{2\omega}{(\omega^2 - 4)} \right]}$$

Step 3 of 4

(b)

Consider the following function.

$$G(t) = 0.5 \sin 10t u(t) \dots\dots (2)$$

It is known that,

$$\begin{aligned}
 G(\omega) &= F(\sin \omega_0 t f(t)) \\
 &= \frac{j}{2} [F(\omega + \omega_0) - F(\omega - \omega_0)]
 \end{aligned}$$

Here,

$$\omega_0 = 10$$

And,

$$\begin{aligned}
 F(\omega) &= F[u(t)] \\
 &= \pi \delta(\omega) + \frac{1}{j\omega}
 \end{aligned}$$

Step 4 of 4

Therefore, the Fourier transform of equation (2).

$$\begin{aligned}
 H(\omega) &= 0.5 \times \frac{j}{2} \left[\pi \delta(\omega+10) + \frac{1}{j(\omega+10)} - \pi \delta(\omega-10) - \frac{1}{j(\omega-10)} \right] \\
 &= \frac{j}{4} \left[\pi \delta(\omega+10) + \frac{1}{j(\omega+10)} - \pi \delta(\omega-10) - \frac{1}{j(\omega-10)} \right] \\
 &= \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] + \frac{j}{4} \left[\frac{1}{j(\omega+10)} - \frac{1}{j(\omega-10)} \right] \\
 &= \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] + \frac{j}{4} \left[\frac{j(\omega-10) - j(\omega+10)}{j(\omega+10)j(\omega-10)} \right] \\
 &= \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] - \frac{1}{4} \left[\frac{j(\omega-10) - j(\omega+10)}{(\omega+10)(\omega-10)} \right] \\
 &= \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] - \frac{1}{4} \left[\frac{-j20}{(\omega+10)(\omega-10)} \right] \\
 G(\omega) &= \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] - \left[\frac{5}{(\omega^2 - 100)} \right]
 \end{aligned}$$

Therefore, the Fourier Transform of $0.5 \sin 10tu(t)$ is

$$\boxed{G(\omega) = \frac{j\pi}{4} [\delta(\omega+10) - \delta(\omega-10)] - \left[\frac{5}{(\omega^2 - 100)} \right]}$$